

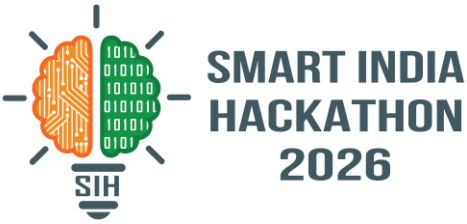
SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – 26169
- **Problem Statement Title** – AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile FSOC Terminals
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** – —
- **Team Name** – ZeroDrift





IDEA TITLE

- ❖ *Proposed Solution (Describe your Idea/Solution/Prototype)*
- *Detailed explanation of the proposed solution*
- *How it addresses the problem*
- *Innovation and uniqueness of the solution*

The eyes and neck of a laser communication terminal

The problem, simply: laser links beam huge data — but only while both terminals point at each other with hair-thin accuracy through shaking, wind and orbital motion. ISRO wants this ‘pointing brain’ proven in software, without costly optics.

Our solution: a virtual test-range — a physics-true sky with stars, clutter, turbulence and a moving blinking beacon — and an AI-assisted tracker that finds it, proves its identity, and steers a virtual camera to keep it centred.

Why it never locks the wrong light: the beacon blinks at an agreed 4 Hz. A star or sun-glint can be brighter, but it cannot blink right — impostors score zero and are rejected.

Smart where it matters: predicts 40 ms ahead to cancel mount delay, rides through blink gaps on a motion model; a self-trained neural verifier double-checks every lock.

Already built: GUI console, standalone app, 73 automated tests, and a zero-install browser demo judges run on their own camera — every number measured, not projected.



1.27 s

acquisition time



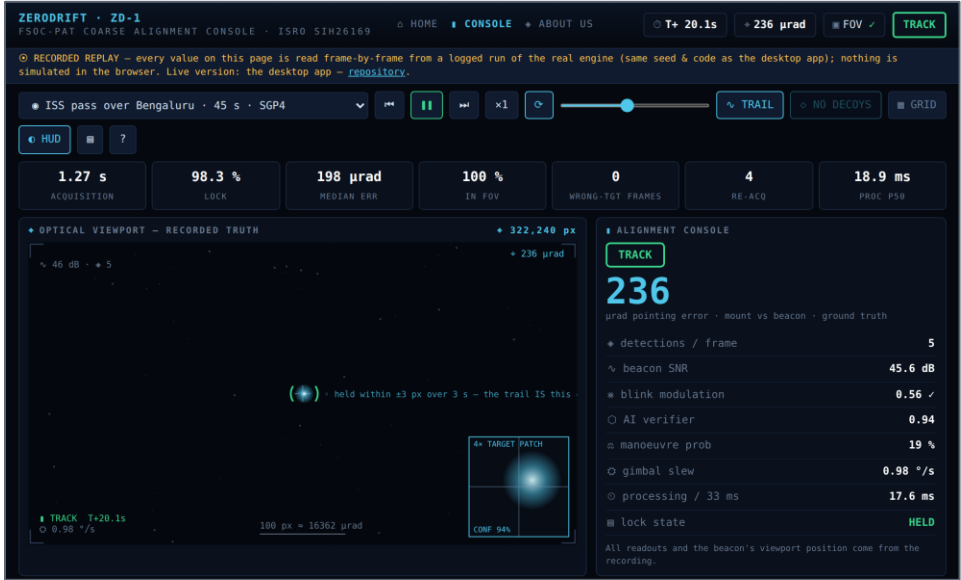
98.3 %

lock retention



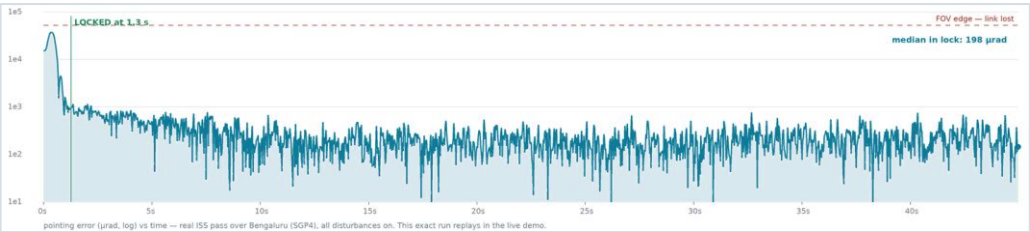
0/64

wrong-target locks



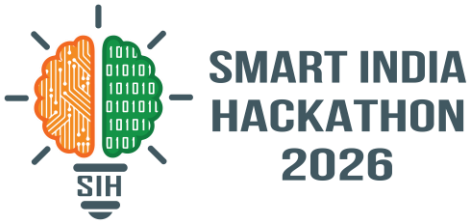
Our console locked — judges can run this tracker on their own camera

Measured: median 198 μ rad in lock · in-FOV 100% · the live-demo run





TECHNICAL APPROACH

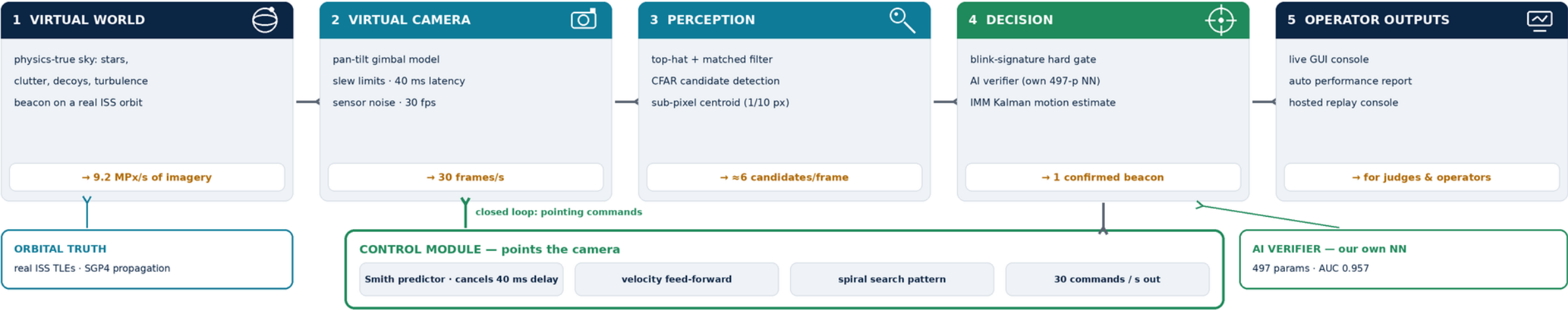


- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

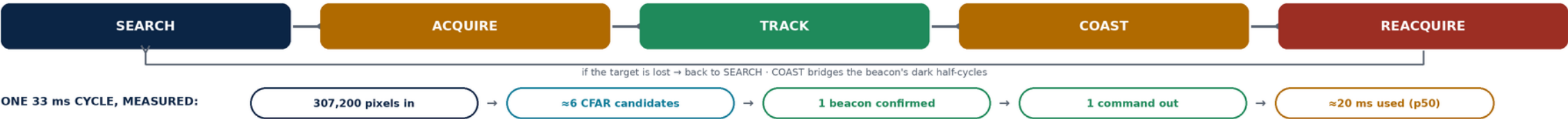
CORE TRACKING PIPELINE



every figure measured — nothing projected



ACQUISITION STATE MACHINE



runs within the frame budget on a normal laptop — no GPU, no cloud, no dataset

BUILT WITH: Python 3.11 NumPy OpenCV PySide6 pyqtgraph SGP4 / real TLEs own neural net pytest 73/ GitHub Actions CI one-click EXE

100% open source · zero licensing cost



FEASIBILITY AND VIABILITY



- *Analysis of the feasibility of the idea*
- *Potential challenges and risks*
- *Strategies for overcoming these challenges*

FEASIBILITY — PROVEN

- Fully working today: 73/73 automated tests pass
- 64-run randomised Monte-Carlo campaign: 100% acquisition, zero wrong locks
- Standalone Windows/Linux app builds in one click; the browser demo needs no install at all
- Tracker uses 8–21 ms of each 33 ms frame on two CPU cores — no GPU, no cloud, no dataset

CHALLENGES & RISKS

- Heavy turbulence dims and shakes the beacon
- Bright decoys / sun glints can mimic the beacon
- Mount latency destabilises fast tracking loops
- Sharp UAV manoeuvres break simple trackers

OUR STRATEGIES

- Coast-through-fade + automatic re-acquisition spiral
- Blink-signature hard gate: a star scores zero — fooled 0 times in 64 runs
- Smith predictor cancels the measured 40 ms delay
- Dual-model IMM filter — two motion models re-weighted every frame catch sharp manoeuvres

Worst case across all 64 randomised runs: slowest acquisition 4.0 s, lowest lock retention 91.6%; even a $0.53\times$ -dim beacon under $1.74\times$ turbulence acquired in 2.8 s. Every number regenerates from one command — fully reproducible.



IMPACT AND BENEFITS



- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

STRATEGIC

Advances India's next-generation space & defence communications — FSOC pointing algorithms developed sovereignly, without importing optical test benches.

ECONOMIC

A laptop replaces lakhs of rupees of cameras, gimbals and optics — the entire develop–test–validate cycle for FSOC pointing runs in pure software, at zero recurring cost.

RESEARCH & EDUCATION

Any student or ISRO lab can develop, test and compare PAT algorithms instantly — reproducible scenarios, honest ground-truth scoring, open source.

SCALABLE & GREEN

Same engine covers satellite, UAV, ground and maritime links. Pure software — no consumables, no lab energy, no e-waste in the development cycle.

Target audience: ISRO / IN-SPACe optical-communication teams, DRDO, IITs & engineering colleges, and India's growing space-tech startups.

OPTICAL LINK AVAILABLE — tracked ISS pass, coarse + modelled fine stage (measured)

WITHOUT coarse alignment



14.7%

WITH ZeroDrift



99.3%

6.8x more usable link time — measured on the same recorded pass, not asserted



RESEARCH AND REFERENCES



• *Details / Links of the reference and research work*

Research foundations

Kaymak et al., “A Survey on Acquisition, Tracking and Pointing Mechanisms for Mobile FSO,” IEEE Comm. Surveys & Tutorials, 2018. [\[link\]](#)

Kaushal & Kaddoum, “Optical Communication in Space: Challenges and Mitigation Techniques,” IEEE Comm. Surveys & Tutorials, 2017. [\[link\]](#)

Blackman & Popoli, Design and Analysis of Modern Tracking Systems — IMM filtering, CFAR detection, track management. [\[link\]](#)

Bar-Shalom et al., Estimation with Applications to Tracking and Navigation — Kalman/IMM theory. [\[link\]](#)

Smith, O.J.M., “Closer Control of Loops with Dead Time” — the Smith predictor, 1957. [\[link\]](#)

Vallado & Crawford, “SGP4 Orbit Determination” — AIAA 2008; real ISS TLE propagation. [\[link\]](#)

Andrews & Phillips, Laser Beam Propagation through Random Media — turbulence & scintillation models. [\[link\]](#)

Our work — open and verifiable

Repository: github.com/pranshukesavmishra/SIH-2026

Live demo: zerodrift-fsoc-pat.netlify.app

Inside: full source (5,000+ lines, documented) · 73 automated tests · 6 validated scenarios · Monte-Carlo logs · technical report · user manual · and /live.html, an in-browser tracker that runs on any laptop or phone camera with nothing installed.

Reproducibility: every figure in this deck regenerates from one command; every random draw is logged and replayable.

Compared with existing approaches

	ZeroDrift	MATLAB sim	HW testbed
Zero hardware cost	✓	✓	✗
Real-time closed loop + live console	✓	✗	✓
Decoy & blink-signature discrimination	✓	✗	✗
Open source · one-command reproducible	✓	✗	✗

Why ZeroDrift wins on this problem

- ✓ All six mandatory deliverables already exist today
- ✓ Judged numbers, honestly measured against hidden ground truth
- ✓ AI where it earns its place — benchmarked against classical methods